Retail Sales Data Analysis

Project Report

1. Introduction

This project is an end-to-end analysis of a retail sales dataset. I used Python and Pandas for data cleaning and exploratory analysis, SQL for deeper analysis, Excel concepts for data handling and lookup work, and Power BI for the final interactive dashboard. The main focus was on sales, orders, customers, products, regions, categories, segments and shipping performance.

The final dataset contains 9,800 rows and 24 columns covering 2015 to 2018. It does not contain a Profit column, so the project focuses on sales and related operational measures.

2. Project Objectives

• Understand and clean the retail sales data.

• Analyze sales performance over time.

• Find product, customer, category and regional patterns.

• Analyze shipping modes and shipping time.

• Use SQL for structured business analysis.

• Build an interactive Power BI dashboard.

• Present the findings in a simple and professional way.

3. Dataset Overview

The main dataset summary is given below.

Metric	Value
Rows	9,800
Columns	24
Years	2015–2018
Unique orders	4,922
Unique customers	793
Unique products	1,861
Regions	4
Categories	3
Segments	3
Ship modes	4
Total sales	$2,261,536.78
Average order value	$459.48
Average ship days	3.96

4. Data Cleaning and Preparation

I checked the dataset structure, data types, missing values and duplicate records. There were no duplicate rows. There were 11 missing Postal Code values. These rows were kept because the missing postal code does not stop the main sales, customer, product or shipping analysis.

The unnecessary Unnamed: 0 column was removed. Date columns were converted to proper date format. Year, Month, Month Name and Year Month fields were prepared for time analysis. Ship Days was used to study delivery time.

5. Python and Exploratory Data Analysis

Python and Pandas were used for the main exploratory analysis. GroupBy, pivot tables, aggregation, percentage calculations, ranking, sorting and date-based analysis were used. The analysis covered sales by segment, category, sub-category, state, region, product and customer.

6. Main Findings

6.1 Overall Sales

Total sales are $2.26 million. There are 4,922 unique orders, 793 unique customers and 1,861 unique products. The average order value is about $459.48.

6.2 Segment Analysis

Consumer has the highest sales among the three segments. Sales were approximately $1.148 million for Consumer, $688,494 for Corporate and $424,982 for Home Office.

6.3 State and Regional Analysis

California has the highest state-level sales, at approximately $446,306. New York follows with about $306,361. Other high-sales states include Texas, Washington, Pennsylvania and Florida. Sales were also compared across the four regions and across categories within each region.

6.4 Category and Sub-Category Analysis

Sales were compared across the three main categories and their sub-categories. Category and segment pivot tables were used to understand the contribution of different groups. These comparisons were later represented in Power BI.

6.5 Product Analysis

Product-level sales were calculated and ranked. The selected product concentration value calculated during the analysis was approximately 10.82%. Top products were represented using the Top 10 Products visual.

6.6 Customer Analysis

Customer-level sales were calculated and ranked. The selected top-customer group contributed about 6.8% of total sales. Top customers were represented using the Top 10 Customers visual.

6.7 Shipping Analysis

Ship Days had 9,800 records, a mean of about 3.96 days, median of 4 days, minimum of 0 days, maximum of 7 days and standard deviation of about 1.75 days. Average shipping time was compared across ship modes using a bar chart. Sales contribution by Ship Mode was shown using a donut chart.

7. SQL Analysis

SQL was used for structured analysis. The work included SELECT, filtering, GROUP BY, aggregate functions, CASE expressions, subqueries, CTEs and window functions. ROW_NUMBER, RANK, DENSE_RANK, NTILE and LAG were studied and used for ranking and previous-row analysis. Running totals and moving averages were also calculated. CTEs were used to break complex queries into smaller steps. Temporary tables, self joins and cross joins were also studied.

8. Excel Concepts

Important Excel concepts covered included Tables, structured references, SUM, AVERAGE, MIN, MAX, COUNT, COUNTA, COUNTBLANK, IF, XLOOKUP and VLOOKUP. These concepts helped build understanding of spreadsheet-based analysis and lookup work.

9. Power BI Dashboard

The final dashboard uses a table named Retail and DAX measures for the main KPIs.

Measure	Purpose
Total Sales	Total sales
Total Orders	Unique orders
Total Customers	Unique customers
Total Products	Unique products
Average Order Value	Average sales per order
Average Ship Days	Average shipping time

9.1 Executive Overview

This page contains KPI cards for total sales, orders, customers, products, average order value and average shipping days. It also contains the monthly sales trend and sales comparisons by category, region and segment.

9.2 Product and Customer Analysis

This page contains Top 10 Products by Sales, Top 10 Customers by Sales, sales by sub-category and sales by category and customer segment.

9.3 Regional and Shipping Analysis

This page contains Top 10 States by Sales, sales by Ship Mode, average shipping time by Ship Mode and regional sales by category.

10. Interactive Filter Panel

A collapsible filter panel was created to keep the dashboard clean. It contains Year, Region, Category and Segment filters. Bookmarks open and close the panel without resetting the selected filters. The slicers are synchronized across all three pages, so a selection made on the main page is applied to the other pages.

11. Key Findings Summary

• Total sales are approximately $2.26 million across 4,922 unique orders.

• Consumer is the largest sales segment.

• California is the highest-sales state.

• Category performance differs across regions and segments.

• The selected top-customer group contributes about 6.8% of total sales.

• The selected product concentration measure is approximately 10.82%.

• Average shipping time is approximately 3.96 days, with a median of 4 days.

• Shipping performance differs by ship mode.

• Monthly and yearly analysis shows changes in sales over time.

12. Conclusion

This project helped me complete a practical end-to-end data analysis workflow. I started with raw retail data, cleaned and checked it, performed analysis using Python and SQL, studied business questions and then built an interactive Power BI dashboard.

The final dashboard gives a simple view of sales performance, customers, products, regions and shipping. The interactive filters allow the same data to be explored from different perspectives.

The main limitation is that the dataset does not contain Profit. Therefore, this project does not make profit or profitability conclusions. If profit data becomes available, profit margin and profitability analysis can be added later.

13. Tools Used

Python, Pandas, NumPy, Matplotlib, Seaborn, MySQL, SQL, Excel, Power BI and DAX.

14. Final Project Deliverables

• Cleaned and analyzed retail dataset

• Python/Jupyter analysis notebook

• SQL analysis queries

• Power BI dashboard (.pbix)

• Three-page interactive dashboard

• Project report

• GitHub repository containing project files and documentation